Electrical measurement of a Metal Oxide Semiconductor Field Effect Transistor (MOSFET)

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Abstract: *I present both the theoretical values as well as the extracted data of the characterization lab findings of the equivalent transmission line resistor circuit and silicon MOSFET electrical characteristics. The I-V characteristics and the fixed resistance value of the line circuit have been calculated and are found to be close to the reference device. The DMOS fabrication process flow has been described, and based on my readings, it has been described how in-situ doping, regrowth of the epitaxial layer, and the use of wide band gap (WBG) materials can significantly enhance the CMOS design technology and the limitations of these changes are also noted. The transfer line characteristics of the resistors suggesting a 730Ω value of the* R_{fix} *resistor is calculated, and its I-V characteristics are also compared. In this experiment, the 2N7000 device is used, and based on its extracted* I_d *Vs* V_d *graph, its* V_{th} *of 1.65 V is extracted, which is within range of the values given in the datasheet. The* I_d *Vs* V_g *graph is also extracted from the* I_d *Vs* V_d *graph using the saturation current formula. The extracted results are then used to identify and describe the specific MOSFET type of the 2N7000 device.*

A. Introduction

The MOSFET electrical parameter characterization has become more important as CMOS technology has advanced to keep scaling down its physical parameters. Electrical characterization involves extracting key device parameters that define the performance, yield, and reliability of MOSFET devices, such as threshold voltage, subthreshold swing, drive current, and leakage current. Accurate measurement and analysis of these parameters is required for process control and validation of simulation models used in circuit design [1]. The variability in electrical parameters that comes with the scaling of these devices has become a critical challenge, affecting circuit delay, power dissipation, and functional yield. Thus, understanding the relationship between these electrical parameters and MOSFET characteristics enables more precise process optimization and more robust circuit design methodologies. In this report, section B goes through the relevant literature, which addresses the scaling of MOSFET based on its electrical characteristics, differences between the conventional NMOS devices and the DMOS device. Section C describes the MOSFET fabrication procedures and also what steps should be taken in order to make a DMOS device. Section D deals with the results of characterization lab findings and compares them with the given 2N7000 MOSFET datasheet, and also in a real-world scenario. Section E concludes the article, and Section F includes the references used in this article.

B. Literature review

The evolution of semiconductor technology can be considered to be based on two key missions: pushing the limits of conventional CMOS scaling and developing specialized wide band gap materials for extreme performance applications [2]. The most important MOSFET parameter to model is the Vth, which is because it can be measured either as drain current or capacitance characteristic using one or several transistors [3]. Nonetheless, numerous extraction schemes based on the properties of the transfer characteristics are highly impacted by source/drain parasitic series resistances and channel mobility degradation [3][4]. Such dependence is not good because it is better to determine an accurate value of Vth without dependence on parasitic effects or mobility changes. Although scaling CMOS devices is facing the basic constraints associated with nanoscale variability, the ability to switch power devices with high voltages and high temperatures requires wide band gap (WBG) materials [5]. A significant challenge for MOSFETs is maintaining a stable, normally off state at high operating temperatures (≤ 200 ºC). This can be resolved via (i) regrowing a thin epitaxial layer over the implanted p-well regions, and (ii) a specialized N_2O-based gate oxidation process. By using these methods, the leakage current decreases notably, i.e. at V_{ds} = 50 mV, the leakage current of 1 μA at 25 ºC is reduced to 0.4 μA at 200 ºC, which confirms that the beneficial mechanism of trap filling dominates also, the epitaxial layer nearly doubled the peak channel mobility and reduced threshold voltage by approximately 2 V compared to a regrowth-free device [5].

Based on geometric efficiency, the channel and JFET components contribute to about 15 -20 % of the total on-resistance optimization. Compared with post-implantation annealing, the epitaxial regrowth is more effective in improving defect reduction in high-voltage devices, especially by using nitrogen implantation. Another way of improving the MOSFET characteristics is by using only a single material for fabrication and reducing the required process steps. One such way is the use of single electron tunnelling devices (SET), which are very fine devices that can be used in quantum computing [6]. An entirely CMOS-compatible fabrication process, by using only silicon, thermally grown SiO2, and p-doped poly-Si in the active region, facilitates the reduction in defects and elimination of metallic oxide instabilities, and commercial viability. SET devices use a lightly boron-doped single crystal Si nanowire on an SOI wafer, controlled by a dual-layer gate configuration, which is used to define and modulate the quantum dot and its tunnel barriers. This method yielded highly controlled, advanced uniform Vth characteristics across different devices and a high dielectric integrity. Although these methods have good performance potential, their high level of fabrication, high cost, and absence of mature fabrication processes render SETs non-viable in large production [7]. At present, the level of technology has reached the node of 3 nm technology, and the further scaling down of the dimensions keeps going on. By the year 2030, TSMC estimates the angstrom level thicknesses of MOSFET layers, and by the year 2037, the industry will have adopted new materials and processes, indicating the start of the post-CMOS era.

Description of MOSFET fabrication procedures

The MOSFET used in the characterization laboratory is the 2N7000 enhancement type N-MOSFET, fabricated using Double-Diffused MOS (DMOS) technology. DMOS is widely used in power electronics due to its capability to support high voltage operation, high current density, and fast switching speeds. An NMOS transistor is fabricated in several major steps, from preparing the silicon substrate to finally making the metal contacts. For the Double-Diffused MOS technology, it is made through two sequential formations of the p-body and n+ source through two lateral diffusions. **First diffusion:** A p-body is diffused into an n-type epitaxial layer. **Second diffusion:** A n+ source is diffused inside the p-body diffusion. Compared to conventional planar NMOS fabrication, to make a DMOS, the main modifications are made in the well/implant scheme, lateral diffusion control, and drift region engineering to enhance breakdown voltage, on-state current, and switching efficiency. Following the methodology outlined in [4], the complete fabrication process can be divided into Front End Of Line (FEOL) and Back End Of Line (BEOL) processing steps.

A. Front End Of Line (FEOL):

a) **Device isolation:** For an N-MOS, the process begins with a lightly doped P-type Si substrate wafer. A thin pad oxide layer of SiO_2 is grown thermally on the Si surface to relieve stress and protect the silicon surface. Then, a layer of Si_3N_4 is deposited by the CVD (Chemical Vapour Deposition) process to act as a hard mask for shallow trench formation. A photoresist is then applied, and photolithography is used to define active regions. After that, dry etching is performed through the nitride and oxide layer all the way into the silicon to form shallow trenches on either side of the active device area. The trenches are then filled with SiO_2 either by CVD or by spin-on-glass so that they are completely filled with insulating material. Lastly, Chemical-mechanical polishing is used to planarize the surface, which removes all the nitride layer and leaves a flat surface with oxide filled trenches isolating the active area. Now, ion implantation of Boron is performed to adjust the p-type doping in the active region. This implantation of Boron makes the concentration appropriate for the desired threshold voltage and the short channel performance.

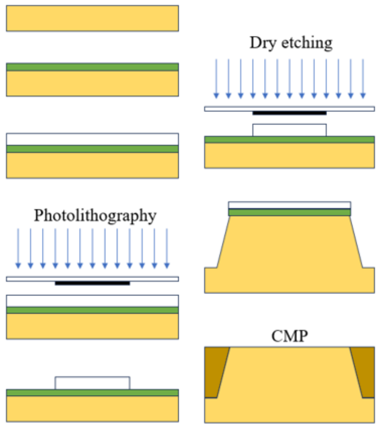

Figure 1 Subthreshold trench isolation (STI)

b) **Gate stack formation:** This step involves the formation of the gate electrode in the MOSFET. An extremely thin layer of gate dielectric, usually SiO_2, is deposited on the exposed active silicon region, using dry thermal oxidation to achieve precise thickness control. This deposited gate oxide defines the length and width of the gate insulator, which greatly influences the threshold voltage and gate capacitance. Now, a poly-Si layer is deposited by the LPCVD process over the entire wafer. Then, the photoresist of the same length as the gate oxide is applied and photolithography is performed to define the gate pattern. Dry etching using SF_6 is now used to remove unwanted poly-Si, leaving a well-defined poly-Si gate pattern on the thin oxide between the trench regions. This is one of the most important steps in the MOS fabrication process, as the gate length ultimately defines the channel length of the transistor.

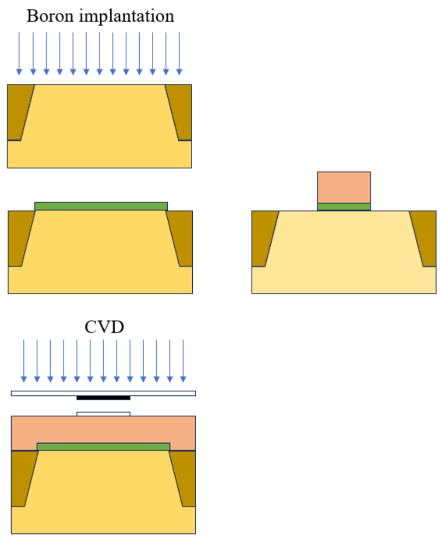

Figure 2 Gate stack formation

c) **Self-aligned source drain formation:** Now that the gate is formed, n-type dopants (phosphorus or arsenic) are implanted into the exposed silicon on either side of the gate to form source and drain regions. As the gate itself acts as a mask, this process does not require any external masks, because of which the source and drain are automatically aligned. This is why this process is called self-aligned source drain formation. After that, high temperature annealing, i.e. 925 to 1000 ºC, is performed to activate the implanted dopants and repair crystal damage.

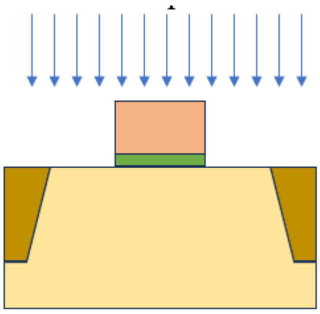

Figure 3 P or As implantation

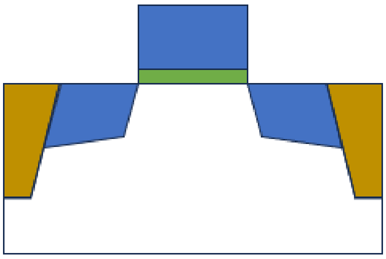

Figure 4 n+ poly-Si gate

d) **SAlicide:** After the above process, a thick layer of SiO_2 is deposited using CVD on the wafer encompassing the gate. Dry etching is now performed across the horizontal surface, enough for it to leave a SiO_2 layer across the gate sidewalls. These sidewall spacers protect the gate edge and help control the junction overlap and series resistance. After this, a thin layer of transition metal such as Ti, Co, or Ni is deposited over the entire wafer and is annealed at about 500-800 ºC. At this temperature, it only reacts with the exposed silicon, i.e. source, drain, and poly-Si to form a low resistance metal silicide. Annealing is used after this to remove all the unreacted metal, leaving self-aligned silicide (SAlicide) on top of the source, drain, and gate, which reduces contact and sheet resistance without requiring extra masks.

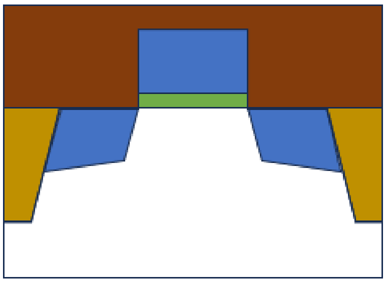

Figure 5 Sidewall spacer deposition

B. Back End Of Line (BEOL):

Metal contact and interconnect formation: A thick layer of SiO_2 is deposited over the entire wafer for passivation, and then CMP is used to planarize the surface. Photolithography is used to open contact holes in the SiO_2 layer all the way to the Silicide surface, and then dry or wet etching is used to remove the unwanted materials. Now the first metal interconnect layer is deposited, and lithography and etching are used to define the device terminals and link them to other circuitry. Additional dielectric and metal layers can then be built on top in a similar manner to complete the full interconnect stack.

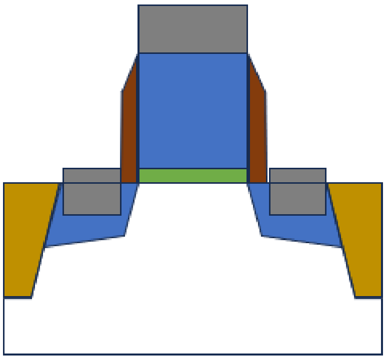

Figure 6 Metal contact deposition

C. DMOS-specific modifications:

A moderately doped n-type epitaxial layer is introduced under the drain and channel to support higher S/D voltages.

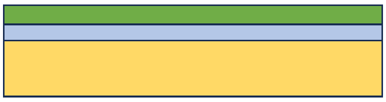

Figure 7 Oxide on the epitaxial layer

- Instead of using only one p-well for the whole NMOS, add a p-type body implant inside the n-drift region near the surface, which will form the channel when inverted.
- Replace the single n+ S/D implant with two lateral diffusions.

Possible modifications in the conventional CMOS process flow:

Modifications in the process flow are usually driven by targets such as improving threshold voltage, drive current, reducing leakage, and increasing switching speed. Some of these process optimizations are:

i. Threshold voltage and subthreshold swing:

It is beneficial to have a threshold voltage as low as possible, but not so much so that it causes high leakage. It can be improved by having a lighter dose of boron in the p-well region. As doping increases, the threshold voltage also increases and reduces the off-state leakage.

The other way of scaling threshold voltage is to slightly increase the gate oxide thickness or to move from pure SiO_2 to a high-k material stack, which can reduce gate leakage at the cost of higher threshold voltage. Other than that, using a thinner stack of gate oxide increases the switching speed.

ii. Mobility, drive current, and channel control:

In a MOSFET, the mobility is a critical parameter that directly influences the drive performance by determining the channel conductivity, drive current, and switching speed. So, to increase mobility, we should reduce the stress and abrupt doping near the channel via subthreshold isolation and well implant conditions. Also, during annealing, we should use a lower temperature activation to preserve effective channel length and improve short-channel control and output resistance.

iii. **Series and contact resistance:**
By increasing the S/D doping and optimizing the silicide formation, it reduces the series and contact resistance, the effect of which is the increase in the on current and a lowering of delay. The sidewall spacers can also be made thinner to increase the on current, but not too thin, as a thicker sidewall spacer improves the breakdown robustness, but at the cost of higher resistance.

iv. **Leakage:**
It can be controlled by scaling the Boron doping near the junction area, which improves standby power and noise margins. Oxide quality can also be improved for cleaner oxidation and reducing the gate tunnelling, which improves static power and subthreshold slope.

v. **Use of epitaxial regrowth:**
This helps in controlling doping profiles, layer thickness, and material composition in specific regions, and is beneficial in making advanced structures like DMOS.

vi. **Use of WBG materials:**
These help the device to operate at higher temperatures, voltages, and switching frequencies than normal CMOS devices. They have lower conduction and switching losses and higher breakdown fields than conventional semiconductors. For example, SiC is such a material.

vii. **Use of in-situ doping:**
This is beneficial because it introduces dopant atoms during the deposition of the semiconductor layer itself, so dopants are incorporated directly into a high quality crystal lattice with minimal damage. This eliminates the lattice defects which are associated with techniques like ion implantation.
Better uniformity in gate length and oxide thickness across the wafer reduces current variations and minimizing dopant diffusion and stress, improves long term reliability and reduces hot carrier and bias temperature instability.

C. Results and discussion

1. Measurement of fixed resistances across the given resistors:

In this experiment, the fixed resistance across the given resistors is being measured using the *two-probe method*. This method appears to be easier to implement, because only two probes need to be manipulated, but the interpretation of the measured data is more difficult. In this method, each contact serves as a current and a voltage probe. So, the total resistance of the DUT is given by

$$R_T = \frac{V}{I} = 2R_C + 2R_W + 2R_{fix} + R_{DUT}$$

Where R_C is the contact resistance and R_{DUT} is the resistance of the device under test. It is clearly impossible to determine the value of the fixed resistance using this arrangement [4].

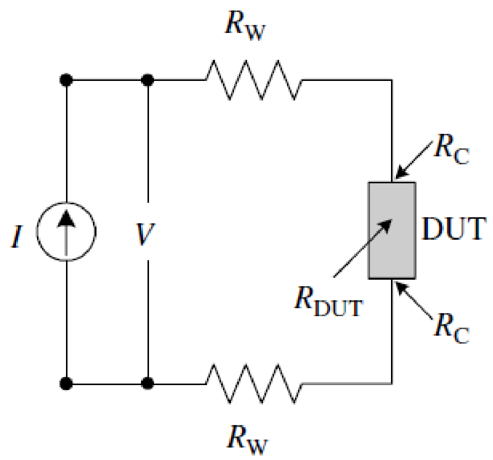

Figure 8 Two-probe resistance measurement [4]

$$R_{fix} = \frac{R_T - 2R_C - 2R_W - R_{DUT}}{2}$$

Table 1 Calculation of fixed resistances

Applied voltage (mV)	R_{DUT} (Ω)	Measured current (µA)	$R_T = \frac{V}{I}$ (Ω)	R_{fix} (Ω)
997.05	1	681.31	1463.4	$731.2 - R_s$
997.04	10	685.28	1454.9	$722.4 - R_s$
997.13	100	635.78	1568.3	$734.1 - R_s$
997	500	496.69	2007.2	$753.6 - R_s$
997.07	1000	374.86	2659.8	$829.9 - R_s$
997.14	5000	93.911	10617.9	$2809 - R_s$
997.15	10000	38.692	25771.4	$7886 - R_s$

Observation:
The R_{fix} is fairly close to each other for the first three rows, but they start to deviate as the R_{DUT} values go up, which suggests that the current has become too low for the meter's precision.
Assuming $R_s = 0$, then $R_{fix} = 730\Omega$ based on the average of the first three rows.

2. I-V Measurement of equivalent transmission (transfer) line resistors circuit using the SMU.

Observation:
Figure 1 represents the equivalent resistance that was measured across the under test resistances. From this graph, the value of the DUT can be measured by calculating the gradient as:

$$V_{applied} = R_{measured} . I_{extracted}$$
$$R_{measured} = R_{DUT} + R_{series}$$

Table 2 Comparison between the DUT resistances and their extracted values

R_{Given} (Ω)	$R_{measured}$ (Ω)
1	1358.75
10	1367.78
100	1457.82
500	1825.97
1000	2353.64
5000	6958.37
10000	11325.15

It can be observed that the measured values from the gradient of the I-V graph for the higher resistances (10k or 5k) are reasonably close to their expected values, but the values for the lower resistors all have significantly higher, and cluster around 1357 Ω.

$$R_T = R_{DUT} + R_S$$
$$R_{DUT} = R_T - R_S$$

It should be noted that these values are extracted using the *two-probe method*, because of which the calculated resistances also include the parasitic series resistances applied because of the measuring equipment. As these parasitic resistances are quite higher than the lower measuring resistances, they are quite noticeable across them. When we take out these series resistances from the calculated values, we will get the true value of R_{DUT}.

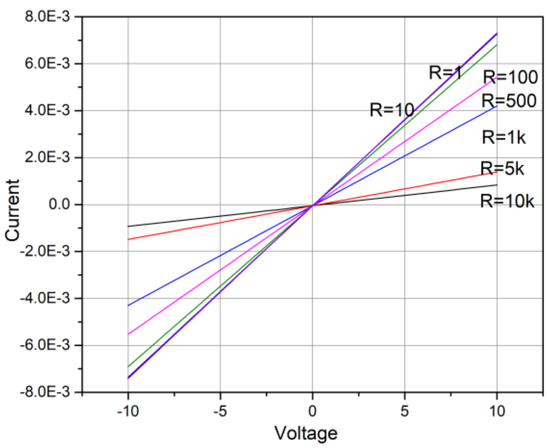

Figure 9 Equivalent resistance plot

3. Measurement of Ids Vs Vds at different fixed Vgs values.

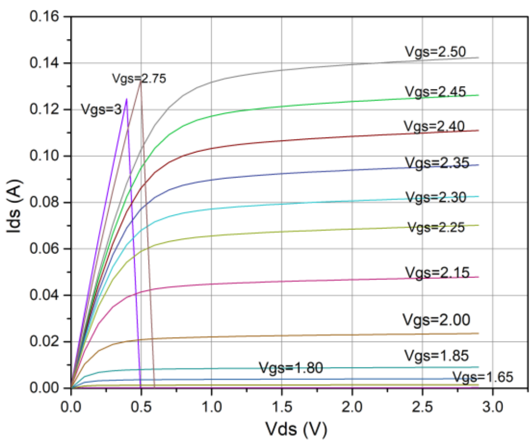

Figure 10 MOSFET Output characteristics

Observation:

This graph represents the output characteristics of the MOSFET. It can be deduced from this graph that the MOSFET starts to get ON at around V_{gs} = 1.65 V, which means that it is the threshold voltage of this MOSFET. As we keep increasing the V_{ds}, more current starts to flow through the MOSFET as higher V_{ds} generates free charges in the channel, but after some time, it gives us a flat output curve as it enters the saturation region. At around V_{ds} = 2.75 V, the current stops flowing as the source to drain voltage reaches the limits of the device.

4. Measurement of Ids Vs Vgs at different fixed Vds values.

From the datasheet of the 2N7000 MOSFET, it is given that the typical value of forward transconductance ($g_m = 320\ mS$)

$$I_{D,Sat} = K(V_{GS} - V_{th})^2 \qquad (i)$$
$$g_m = \frac{dI_D}{dV_{GS}} \qquad (ii)$$
$$K = \frac{g_m}{2(V_{GS} - V_{th})} \qquad (iii)$$

Therefore, after inputting the calculation values

$$K = \frac{320 * 10^{-3}}{2(3 - 1.65)}$$
$$K = 0.118$$

The values of the gate voltages V_{GS} can be extracted from Figure 10.

Now, inputting the value of K in equation (i), the saturation current across the different values of V_{GS} is calculated:

Threshold voltage can be calculated by finding the point where the I_{ds} Vs V_{gs} graph gives the value of zero, as this is the point where the gate voltage is equal to the threshold voltage and from equation (i), the saturation current becomes zero.

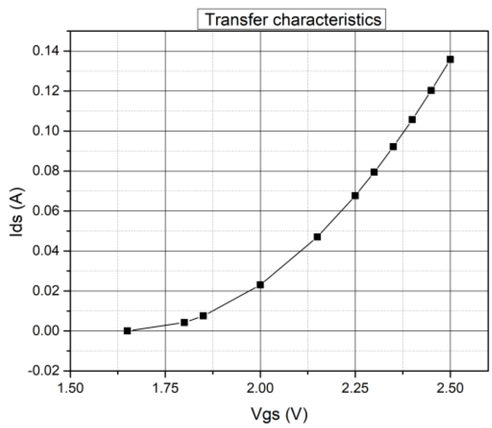

Figure 11 MOSFET Transfer characteristics

Table 1 Saturation current across different values of gate voltage

V_{GS} (V)	$I_{D,Sat}$ (A)
1.65	0
1.80	0.002655

1.85	0.00472
2.00	0.014455
2.15	0.0295
2.25	0.04248
2.30	0.049855
2.35	0.05782
2.40	0.066375
2.45	0.07552
2.50	0.085255
2.75	0.14278
3.00	0.215055

Observation:
It is given in the datasheet that the typical threshold voltage of the 2N7000 MOSFET is 2.1 V, but the practical value we are getting from the lab is around 1.65 V. These can be the following reasons for this difference:

- **Datasheet specification:**

The measured value falls well within the minimum, typical, and maximum range for threshold voltage. This value can minutely change because of the process variations, how old the MOSFET is, and how it is being tested.

- **Operating conditions:**

Threshold voltage decreases as the temperature increases, so if the temperature during measurement exceeds the typical value of 25 °C, the threshold voltage can get lower.
Based on the measured results, it can be deduced that 2N7000 is well suited for low-power switching applications as it has a lower V_{th} compared to other devices. It has a significantly low R_{ON} value, which is great for a smooth flow of current. Overall, it is a great device to be used in power electronics.

5. Measurement of the MOSFETs drain-source ON resistance (Rds).

The ON-resistance of a MOSFET is the resistance between its drain and source terminals when the device is conducting and fully switched on, which is measured at a specified V_{gs}. A lower R_{ON} is desirable as it results in low power loss as heat and provides more efficient current flow.

It can be extracted from the datasheet that the Ron for the 2N7000 MOSFET operating at $V_{gs} = 4.5\,V$, $I_{ds} = 75\,mA$ gives a typical value of $R_{ON} = 1.8\,\Omega$ and a maximum value of $R_{ON} = 5.3\,\Omega$

Table 2 Extracted values of on-resistances at different gate to source voltage

V_{gs} (V)	V_{ds} (mV)	I_{ds} (mA)	R_{ON} (Ω)
3	69.5	11	6.31
	240	30	8
	558	100	21.3
3.5	64.5	11	5.863
	160	30	5.33
	811	100	8.11
4	49.61	11	4.51
	133.61	30	4.45
	524	100	5.24

Table 4 shows the extracted results of the R_{ON}, which are strikingly similar to the data given in the datasheet.

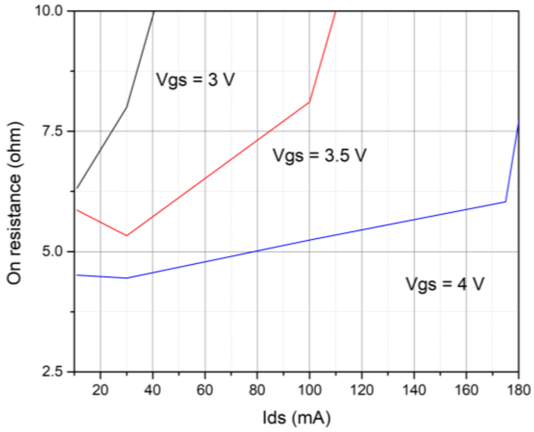

Figure 12 MOSFET ON resistance

D. Conclusion

To conclude, this report has discussed the effect of scaling down MOSFET parameters on the performance and variability of devices. It also described the procedures of making an NMOSFET and what changes are needed in that procedure to make a DMOS device. All these reveal both the possible benefits of these emerging technologies and the shortcomings and challenges that they are currently experiencing. The characterization lab results and analysis have also been given and discussed in detail.

E. References

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